AW-CAR / PROTOTYPE V2 / A3 WIRING

Purple: Brain | Blue: Motion | Green: FPGA | [PLAN]: future wiring, firmware not enabled

J1
BENCH <=8.4V / CC
+8.4V
GND

U15
DSN6000AUD / 5V
IN+
IN-
OUT+_=_5.0V
OUT-
5V_LOGIC

JP1
RUN / USB
RUN
5V

JP2
RUN / USB
RUN
5V

U16
BUCK AMP 5V / TBD
IN+
IN-
OUT+_=_5.0V
OUT-

JP3
RUN / USB
RUN
5V

U1
ESP32-S3 BRAIN
5V_JP1
3V3
5V_BRAIN
3V3_BRAIN
GPIO18_/_UART_RX
GPIO17_/_UART_TX
BM_MOTION_TX
BM_BRAIN_TX
GPIO4_/_MIC_WS
GPIO5_/_MIC_BCLK
GPIO6_/_MIC_SD
MIC_SD
MIC_WS
MIC_BCLK
GPIO7_/_AMP_DIN
GPIO15_/_AMP_BCLK
GPIO16_/_AMP_LRC
AMP_DIN
AMP_BCLK
AMP_LRC
GPIO41_/_OLED_SDA
GPIO42_/_OLED_SCL
OLED_SDA
OLED_SCL
GND

U2
ESP32-S3 MOTION
5V_JP2
3V3
5V_MOTION
3V3_MOTION
GPIO14_/_UART_RX
GPIO13_/_UART_TX
GPIO16_/_MISO
SPI_MISO
GPIO15_/_MOSI
GPIO4_/_SCLK
GPIO5_/_CS_n
SPI_MOSI
SPI_SCLK
SPI_CS_N
GPIO11_/_SDA_[P]
GPIO12_/_SCL_[P]
SENSOR_SDA
SENSOR_SCL
GPIO8_/_EDGE_L_[P]
GPIO9_/_EDGE_R_[P]
EDGE_L_DO
EDGE_R_DO
GPIO10_/_STBY
STBY_COMMAND
GND

U3
TANG NANO 9K
5V_JP3
3V3
5V_TANG
3V3_TANG
P26_/_MISO
P25_/_MOSI
P29_/_SCLK
P30_/_CS_n
P35_/_LEFT_IN1
P40_/_LEFT_IN2
P27_/_ENC_L_[P]
P41_/_RIGHT_IN1
P42_/_RIGHT_IN2
P28_/_ENC_R
ENC_L_3V3
ENC_R_3V3
L_IN1
L_IN2
R_IN1
R_IN2
GND

U4
DRV8833 LEFT
VM_8V4
AIN1
BIN1
AIN2
BIN2
STBY/nSLEEP
GND1
GND2
AO1
AO2
BO1
BO2
L_AO1
L_AO2
L_BO1
L_BO2
STBY_L
M1
+ TT LF
-
M2
+ TT LR
-

U5
DRV8833 RIGHT
AIN1
BIN1
AIN2
BIN2
STBY/nSLEEP
GND1
GND2
AO1
AO2
BO1
BO2
R_AO1
R_AO2
R_BO1
R_BO2
STBY_R
M3
+ TT RF
-
M4
+ TT RR
-

R1
1k STBY L
1
2

R2
1k STBY R
1
2

U9
LEVEL SHIFT 4CH
HV_5V
LV_3V3
HV1_[PLAN]
LV1_[PLAN]
HV2
LV2
HV3_/_NC
LV3_/_NC
HV4_/_NC
LV4_/_NC
GND1
GND2

INMP441
3V3
SCK/BCLK
WS
SD
L/R_=_LEFT
GND

MAX98357A
5V_AMP
DIN
BCLK
LRC
GAIN_/_OPEN
SD_/_OPEN
SPK+
SPK-
SPK_P
SPK_N
GND

OLED I2C
3V3
SDA
SCL
GND

U11
MPU6050 [PLAN]
3V3
SDA_/_0x68
SCL
AD0_=_0
INT_/_NC
XDA_/_NC
XCL_/_NC
GND

U12
VL53L0X [PLAN]
3V3
SDA_/_0x29
SCL
XSHUT_/_NC
GPIO1_/_NC
GND

U13
TCRT5000 L [PLAN]
3V3
DO_/_EDGE_L
AO_/_NC
GND

U14
TCRT5000 R [PLAN]
3V3
DO_/_EDGE_R
AO_/_NC
GND

U17
ENC 1 OUT L [PLAN]
3V3
OUT
GND
ENC_L_5V

U10
ENC 1 OUT R
3V3
OUT
GND
ENC_R_5V

LS1
4 ohm / 3 W
SPK+
SPK-

ENC L: change P27 pull to NONE first. TCRT DO / sensor I2C: 0..3V3. Planned pins not enabled.

RUN: close JP1/2/3 only with board USB unplugged. DEBUG: open that board jumper BEFORE USB.

GPIO17/18 Motion UNUSED. Pxx = PACKAGE pin. System input <=8.4V: VM bypasses the DSN.

DSN 5V: NO-LOAD verified only. AMP buck/load/filters TBD. SPK- is NOT GND.

Draft: no PCB, no motor/power acceptance

AW-Car

Sheet: /
File: simplev1.kicad_sch

Title: AW-Car prototype v2 - Module wiring

Size: A3	Date: 2026-11-11	**Rev: A3-2**
KiCad E.D.A. 10.0.6		Id: 1/1